

Introduction and housekeeping

Using Networks to Study Microbes

Alberto Santos & Yesid Cuesta-Astroz



Outline

- Who are you?
- Introduction to the course
- Course schedule
- Tools
- Contact

Who are you?

Alberto Santos - Associate Professor and Head of the Informatics Platform at the Danish Technical University (DTU) - Denmark

- Research Group Lead **Multi-omics Network Analytics** (MoNA )

- **Multi-omics**

Using multimodal data to have a comprehensive view on (micro) biology problems

- **Network**

Exploiting graphs to structure, represent, integrate and analyse data

- **Analytics**

Applying Data Science and Machine Learning to answer complex biological questions

Who are you?

Yesid Cuesta-Astroz - Associate professor



- Associate Professor / Universidad de Antioquia / Escuela de Microbiología
 - Pathogen genomics and bioinformatics
 - Working on the study of the biological diversity of microorganisms from omics approaches.
 - Modelling the host-pathogen interactions using protein-protein interaction networks.

Who are you?

Introduction

Introduction to the Course

Networks to Study Microbes

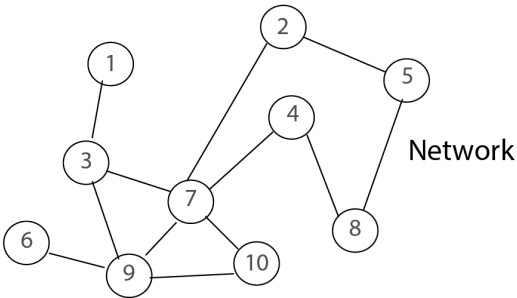
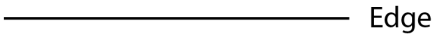
- Technological advances are making **large-scale omics datasets** available to **study microbes and microbial communities**
- We need **computational methods** to **model and understand** these complex data
- This course will provide an overview of **omics data types** and **data resources** available and how to **integrate, analyse and interpret** these data with **networks**

Introduction to the Course

What are networks?

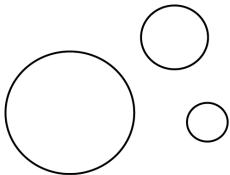
- **Data structures** of components (**nodes**) connected by relationships (**edges**)
- These structures allow:
 - Quick **integration** of **heterogeneous data** based on relationships
 - **Graph theory** methods can be used to **analyse** and **interpret** data, e.g., topological properties can be used to explain:
 - The possible **role** of specific components
 - The **flow** of information
 - The **robustness** of the system
- **Visualize** data

Networks

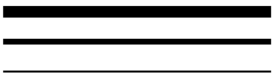


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0	1	1	1	0	0	0	0	1	1
0	0	0	1	1	0	0	0	0	0
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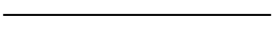
Adjacency matrix



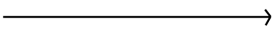
weighted nodes (size)



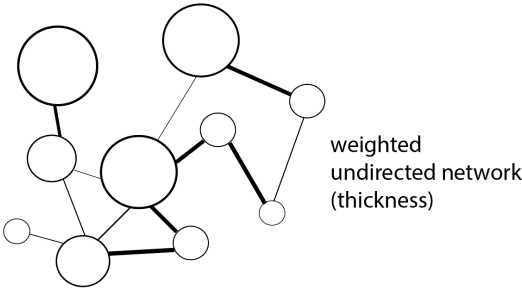
weighted edges (thickness)



undirected edge

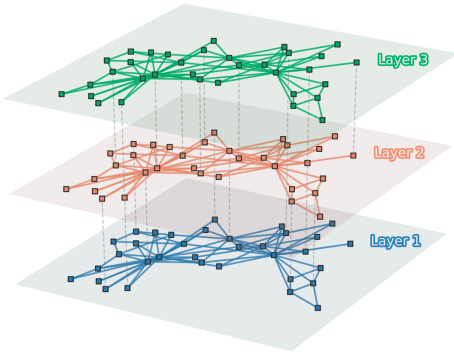
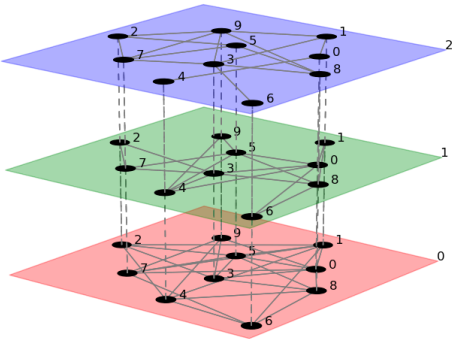


directed edge



0	0	%	0	0	0	0	0	0	0
0	0	0	0	%	0	%	0	0	0
%	0	0	0	0	0	%	0	%	0
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0	%	0	0	0	0	0	%	0	0
0	0	0	0	0	0	0	0	%	0
0	%	%	%	0	0	0	%	%	%
0	0	%	%	%	0	0	0	0	0
0	0	%	0	0	%	0	0	%	%
0	0	0	0	0	0	%	0	%	0

Weighted adjacency matrix



Schedule

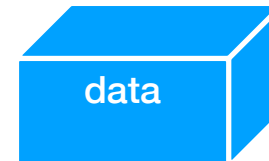
Time	DAY 1	DAY 2	DAY 3	DAY 4
9:00-9:45	Introduction and Housekeeping	Introduction to Python I	Analysing Networks I	Graphical User Interface
9:45-10:30	An Omics View on Microbes I	Introduction to Python II	Analysing Networks II	Graphical User Interface
10:30-11:00	Coffee break	Coffee break	Coffee break	Coffee break
11:00-11:45	An Omics View on Microbes II	Working with Data in Python I	Visualising Networks I	Invited Speaker
11:45-12:30	Open Science	Working with Data in Python II	Network Exercises	Introduction to Team Projects
12:30-14:00	Lunch	Lunch	Lunch	Lunch
14:00-14:45	Network Biology	Databases & Data Annotation	Visualising Networks II - Gephi	Team Project
14:45-15:30	Network Biology journal club	Visualizing Data in Python	Visualising Networks II - Cytoscape	Team Project
15:30-16:00	Coffee break	Coffee break	Coffee break	Coffee break
16:00-16:30	Network Biology journal club	Visualizing Data in Python	Cytoscape/Gephi exercise	Team Project
16:30-17:00	Recap and Q & A	Recap and Q & A	Recap and Q & A	Team Project Presentations and Q&A

Tools

Course Website

<https://github.com/Multiomics-Analytics-Group/networks> to study microbes

- We use GitHub to maintain the contents of the course
- **GitHub** is a framework generally used to maintain and version control software development projects
- **Version control** is a system that records changes to a file or a set of files over time and who made them so that you can recall specific versions later

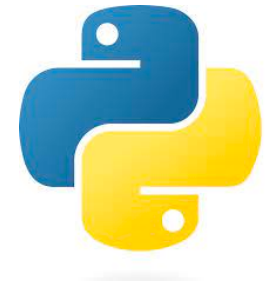


<https://en.wikipedia.org/wiki/GitHub>

<https://git-scm.com/book/en/v2/Getting-Started-About-Version-Control>

Basic Programming

Python



- Python is a **high-level, general-purpose programming language**
- It is a good language for both **beginners** and **advanced programmers**:
 - Its design philosophy **emphasises code readability**
 - Many **learning resources** available
 - **Large community** behind (i.e. stackoverflow for questions)
 - **Excellent Libraries** available, especially for data science

Jupyter Notebooks

<https://jupyter.org/>



- **Web-based development environment** for **creating, running** and **sharing Python** (and other languages) code
- A **notebook** is an interactive document that combines **live code, equations, text or markdown**, and **visualisations** (output of your code)
- Notebooks are divided into **cells** that **run sequentially!** (Need to pay attention)
- It **requires** having **Python installed** on your local machine



Colab Notebooks

<https://research.google.com/colaboratory/faq.html>

- Google Colab is based on **Jupyter Notebook** open source project **hosted on Google's servers**
- Advantages:
 - Requires **no setup** to use (no python installation)
 - Provides **free** access to **computing resources** on Google's servers including GPUs
 - **Notebooks** can be **shared** just as you would with Google Docs or Sheets.
 - You can **import** existing **Jupyter notebooks**
- **Own data and notebooks** need to be accessed through **Google Drive** — Need Google account

Gephi

<https://gephi.org/>



- **Open-source software** for **visualising** and **analysing** large networks graphs
- Used for any kind of networks, especially large networks — good performance
- **Extendable** using Plugins — <https://gephi.org/plugins>

Cytoscape

<https://cytoscape.org/>



- An open source **software platform** for **visualising** and **analysing** complex **networks**
- Used for any kind of networks but **specialised** on **biological domains**:
e.g, Molecular interaction networks and biological pathways and integrating these networks with annotations, gene expression profiles and other state data.
- **Additional features** are available as freely available **Apps** (<https://apps.cytoscape.org/>)

Data Literacy

Data is like garbage. You'd better know what you are going to do with it before you collect it. Mark Twain

What is it?

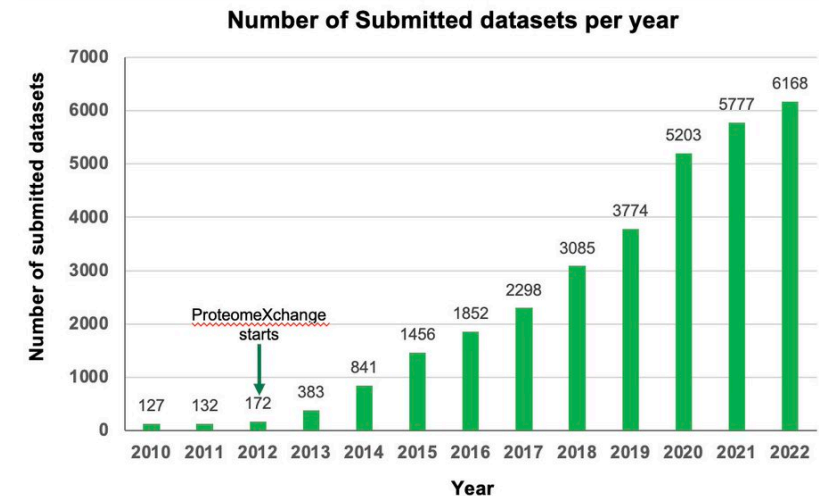
Data as a universal language

- The ability to **read, analyse** and **communicate** data effectively
- Includes also understanding **data sources** and **formats**, and the **analytical methods** to transform them into results
- Requires skills and knowledge in **data analysis, statistics, data visualisation**, and **data management**.
- It does not mean that we all need to be **Data Scientist ...**
- **but** we all need certain skills to extract value from data:
 - **Non-technical**: interpretation, critical thinking, communication, evaluation
 - **Technical**: analysis, visualisation, programming, management

Why?

Data! Data! Data! Evidence to support our research

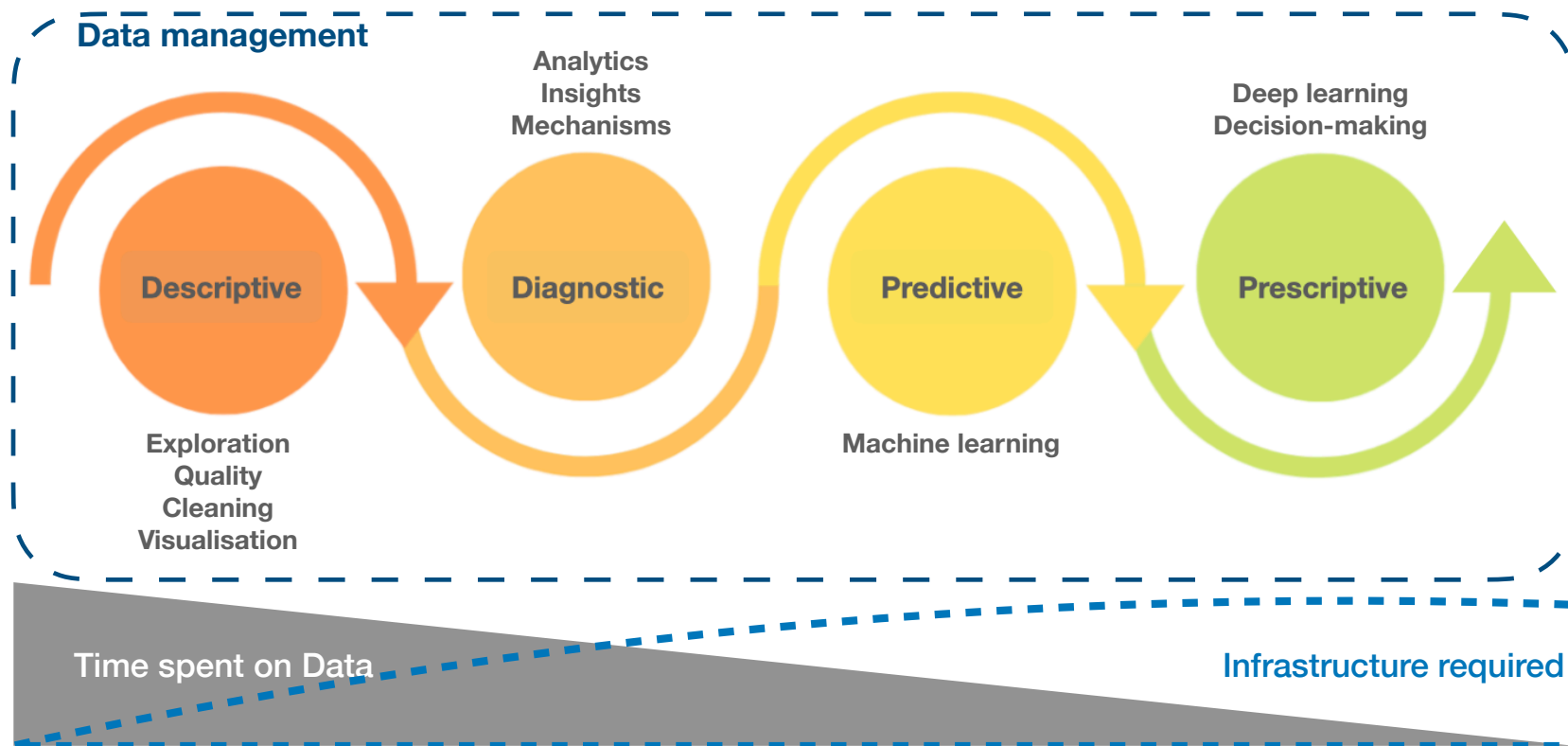
- We **generate data** everyday: metadata, raw, processed, clean, derived results
- **Growing number** of **resources** and **datasets** available
- The data can have **multiple purposes** — reusability (FAIR)
- Benefits:
 - Better **understanding** of data
 - Extract **data-driven** insights
 - **Freedom** to create, format and analyse
 - **Meaningful** communication with data
 - Manage your data **sustainably**



Our Data Journey

Be Data Literate: The Data Literacy Skills Everyone Needs To Succeed. Jordan Morrow

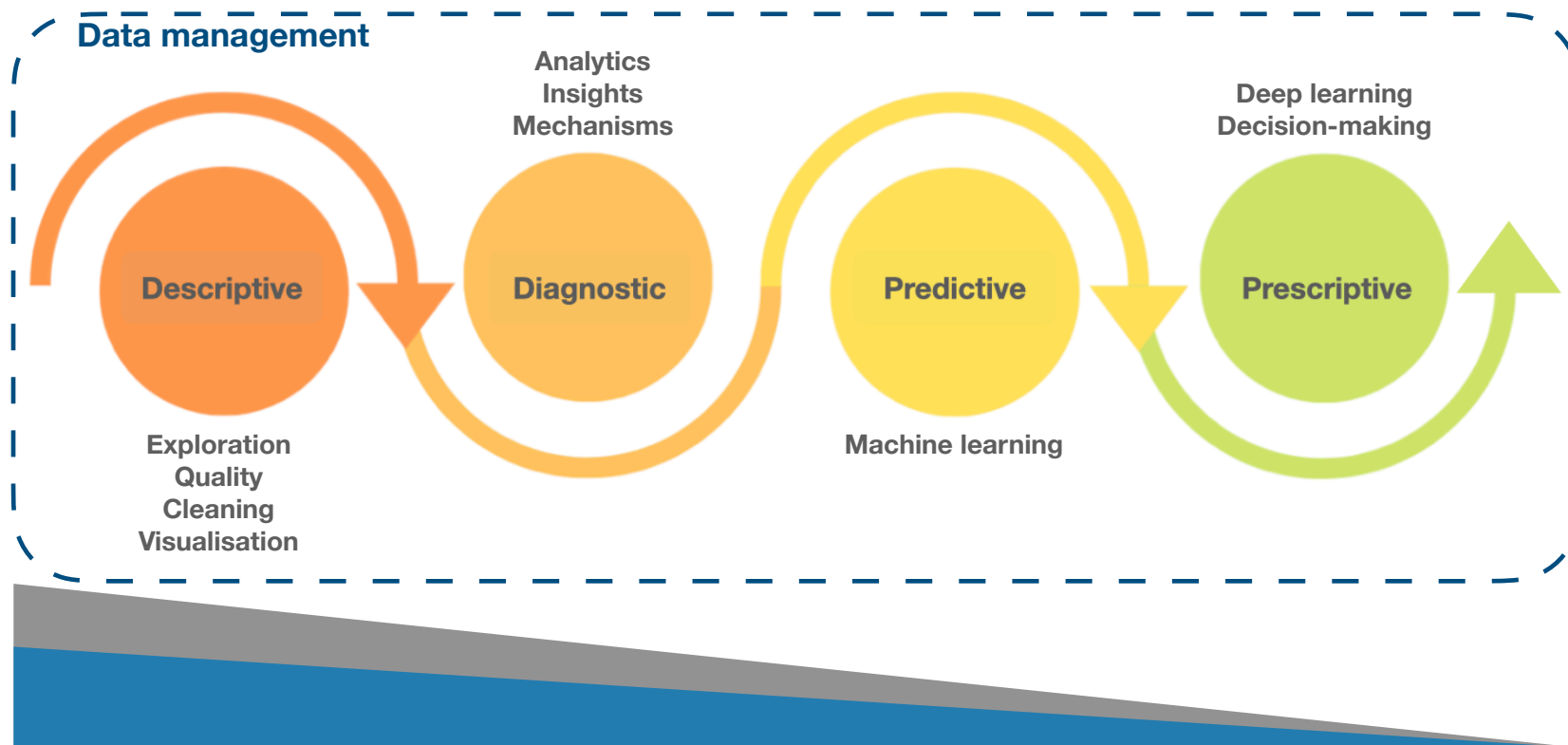
4 Levels of Data Literacy



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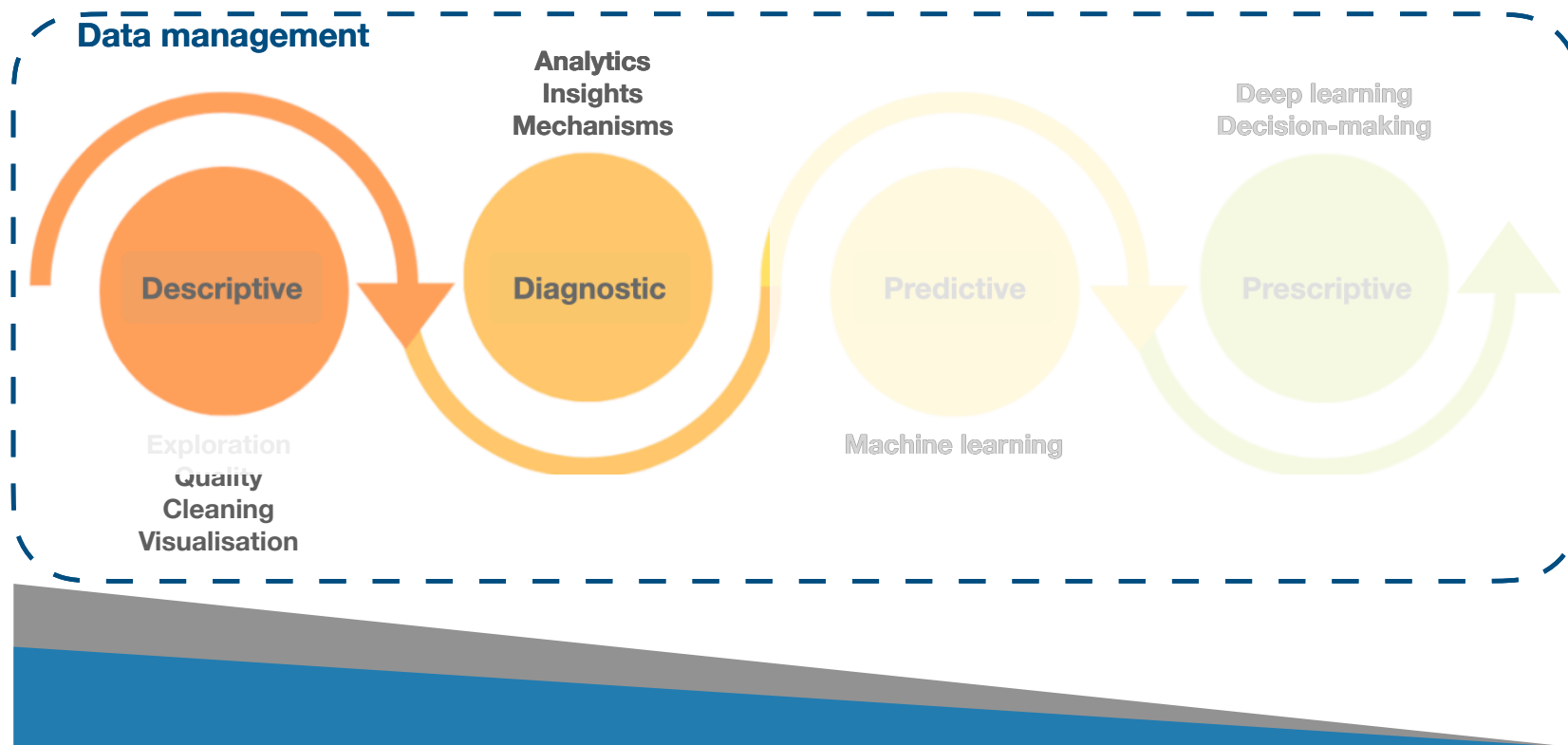
4 Levels of Data Literacy



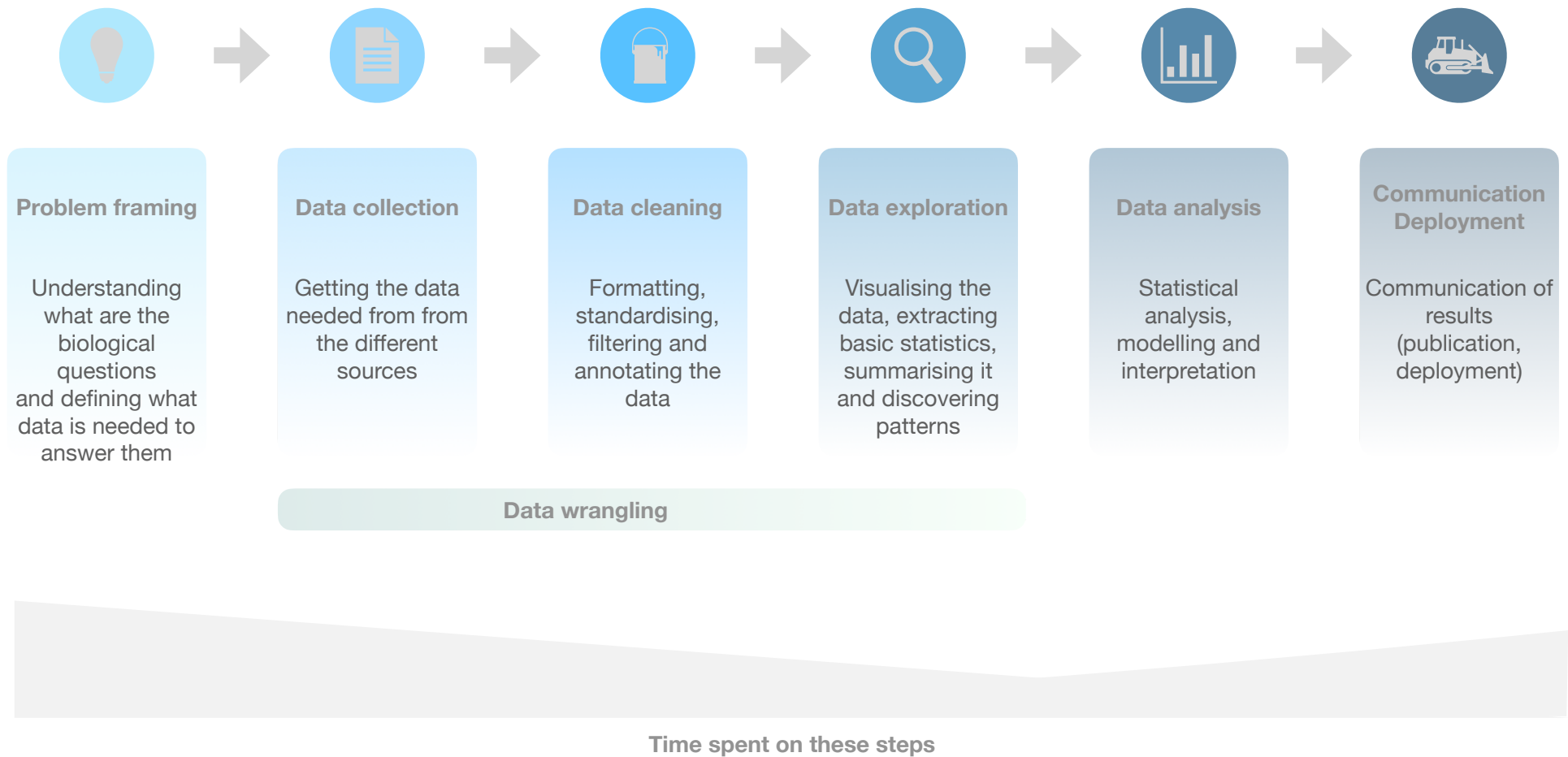
Our Data Journey

Be Data Literate: The Data Literacy Skills Everyone Needs To Succeed. Jordan Morrow

4 Levels of Data Literacy



The Data Science Process



Relevant Links

Extra readings and resources

- **Network biology: understanding the cell's functional organisation.** Albert-László Barabási & Zoltán N. Oltvai. Nature Reviews Genetics 2004 (<https://www.nature.com/articles/nrg1272>)
- **Network analysis of protein interaction data** — Online tutorial — EMBL EBI (<https://www.ebi.ac.uk/training/online/courses/network-analysis-of-protein-interaction-data-an-introduction/>)
- **Network Biology: A short introduction to the core concepts** <https://www.youtube.com/watch?v=H1bGk8PGvf8>
- **Network Visualization: a short introduction to the core concepts of network layout and clustering** <https://www.youtube.com/watch?v=OunX7ISRbgA>
- **A Guide to Conquer the Biological Network Era Using Graph Theory.** Mikaela Koutrouli, Evangelos Karatzas, David Paez-Espino, Georgios A. Pavlopoulos. Frontiers in Bioengineering and Biotechnology 2020 (<https://www.frontiersin.org/articles/10.3389/fbioe.2020.00034/full>)
- **Python for Beginners** <https://www.python.org/about/gettingstarted/>
- **How to Use Jupyter Notebook: A Beginner's Tutorial** <https://www.dataquest.io/blog/jupyter-notebook-tutorial/>
- **Getting started with Python and Jupyter Notebooks** <https://colab.research.google.com/github/jckantor/CBE30338/blob/master/docs/01.01-Getting-Started-with-Python-and-Jupyter-Notebooks.ipynb>

Course Datasets

Data wrangling:

Proteome allocations change linearly with the specific growth rate of *Saccharomyces cerevisiae* under glucose limitation

[Jianye Xia](#), [Benjamin J. Sánchez](#), [Yu Chen](#), [Kate Campbell](#), [Sergo Kasvandik](#) & [Jens Nielsen](#) ✉

[Nature Communications](#) 13, Article number: 2819 (2022) | [Cite this article](#)



Download historical data (to 14 December 2020) on the daily number of new reported COVID-19 cases and deaths worldwide **Archived**

14 Dec 2020

Data visualisation:

High-quality binary protein interaction map of the yeast interactome network

Haiyuan Yu ¹, Pascal Braun, Muhammed A Yildirim, Irma Lemmens, Kavitha Venkatesan, Julie Sahalie, Tomoko Hirozane-Kishikawa, Fana Gebreab, Na Li, Nicolas Simonis, Tong Hao, Jean-François Rual, Amélie Dricot, Alexei Vazquez, Ryan R Murray, Christophe Simon, Leah Tardivo, Stanley Tam, Nenad Svrzikapa, Changyu Fan, Anne-Sophie de Smet, Adriana Motyl, Michael E Hudson, Juyong Park, Xiaofeng Xin, Michael E Cusick, Troy Moore, Charlie Boone, Michael Snyder, Frederick P Roth, Albert-László Barabási, Jan Tavernier, David E Hill, Marc Vidal

Network visualisation:

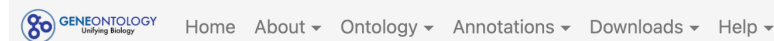
Transcriptional and proteomic insights into the host response in fatal COVID-19 cases

[Meng Wu](#), [Yaobing Chen](#), [Han Xia](#) ^{1,2}, and [Dennis L. Kasper](#) ¹ ✉ [Authors Info & Affiliations](#)

Contributed by Dennis L. Kasper, September 22, 2020 (sent for review August 26, 2020; reviewed by Arturo Casadevall and Robert W. Finberg)

October 20, 2020 | 117 (45) 28336-28343 | <https://doi.org/10.1073/pnas.2018030117>

Data annotation:



[Downloads](#) / [Download Annotations](#)

Download Annotations for 2024-11-03 release



Programmatic access

UniProt provides several application programming interfaces (APIs) to query and access its data programmatically:

UniProt website REST API

What: RESTful URLs that can be bookmarked, linked and used in programs for all entries, queries and tools available through this website. Data is available in all formats provided on the website, e.g. text, XML, RDF, FASTA, GFF, tab-separated for UniProtKB protein data.

Why: Access data and tools from the UniProt website with any programming language.

Documentation: <https://www.uniprot.org/help/api>

Proteins REST API

What: Extended REST API with a service providing [genomic coordinates](#) of UniProtKB sequences, and other services providing annotations imported and mapped from Large Scale data Sources (LSS), such as 1000Genomes, ExAC, PeptideAtlas, and HPA via the [variation](#), [proteomics](#), and [antigen](#) services.

Why: Access data sets mapped to UniProt and integrated through a single service.

Documentation: <https://www.ebi.ac.uk/proteins/api/doc/>